

9.2 Resistance

Question Paper

Course	CIEA Level Physics
Section	9. Electricity
Topic	9.2 Resistance
Difficulty	Hard

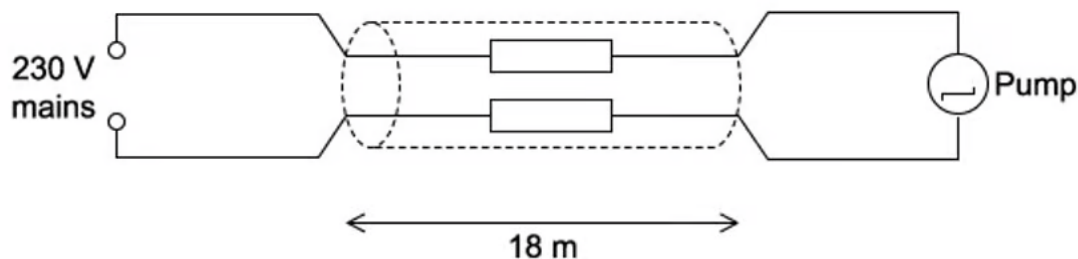
Time allowed: 20

Score: /10

Percentage: /100

Question 1

The diagram shows an electric pump for a garden fountain connected by an 18 m cable to a 230 V mains electrical supply.



The performance of the pump is acceptable if the potential difference (p.d.) across it is at least 218 V. The current through it is then 0.83 A.

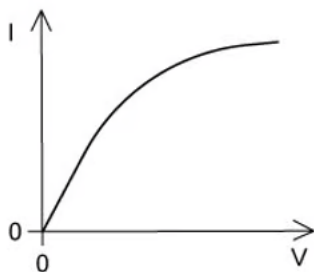
What is the maximum resistance per meter of each of the two wires in the cable if the pump is to perform acceptably?

- A. $0.40 \, \Omega \text{ m}^{-1}$
- B. $0.80 \, \Omega \text{ m}^{-1}$
- C. $1.3 \, \Omega \text{ m}^{-1}$
- D. $1.4 \, \Omega \text{ m}^{-1}$

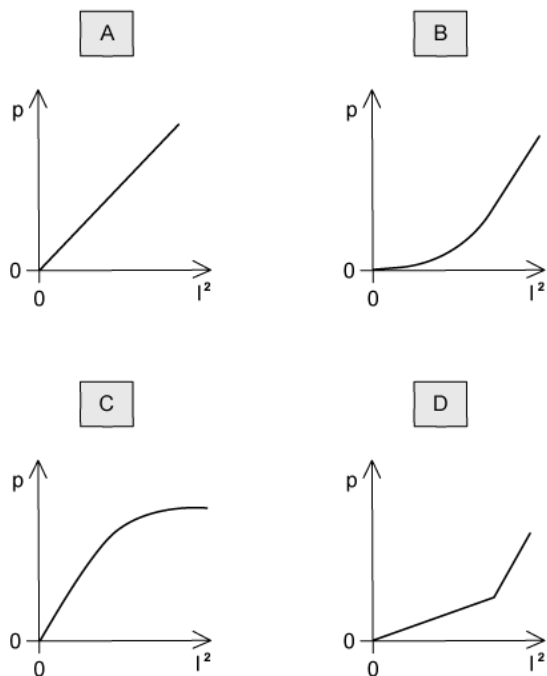
[1 mark]

Question 2

The graph below shows the variation of the current I through a lamp with potential difference V across it.



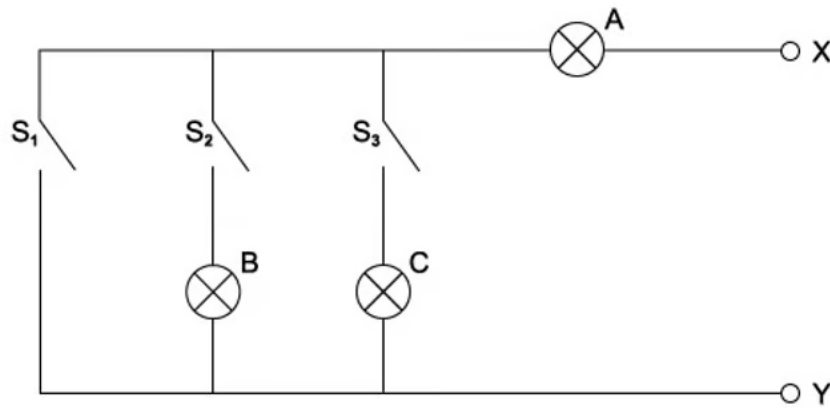
Which of the following graphs best represents the variation of power P dissipated in the same lamp with I^2 ?



[1 mark]

Question 3

Circuit contains three identical lamps **A**, **B** and **C**. Circuit also contains three switches, S_1 , S_2 and S_3 , as shown.



One of the lamps is faulty. In order to detect the fault, an ohm-meter is connected between terminals X and Y.

When measuring resistance, the ohm-meter causes negligible current in the circuit. Readings of the ohm-meter for different switch positions are shown in the table.

Switch			Meter reading / Ω
S_1	S_2	S_3	
open	open	open	∞
closed	open	open	15
open	closed	open	30
open	closed	closed	15

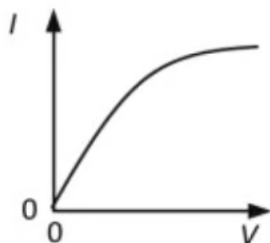
Which lamps are faulty?

- A. A & B
- B. B only
- C. B & C
- D. C only

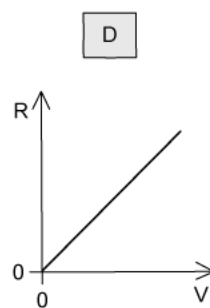
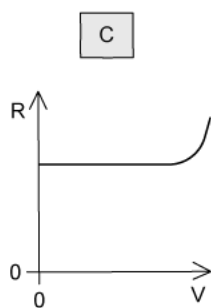
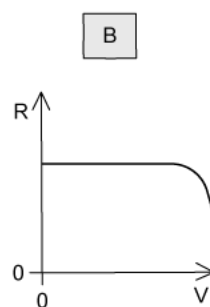
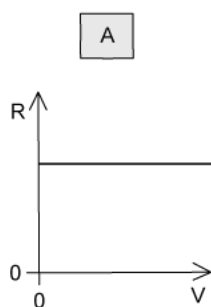
[1 mark]

Question 4

The current flowing through a component varies with the potential difference V across it as shown.



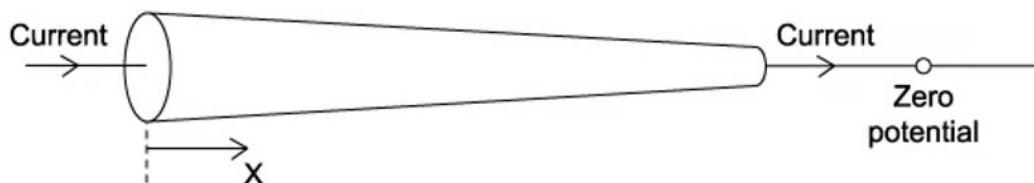
Which graph best represents how the resistance R varies with V ?



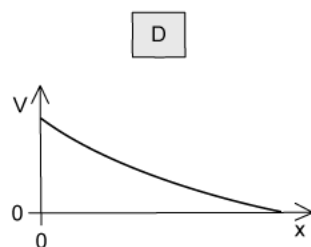
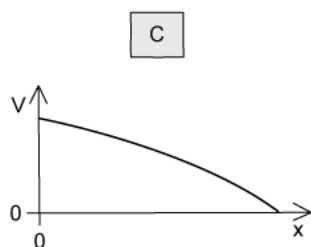
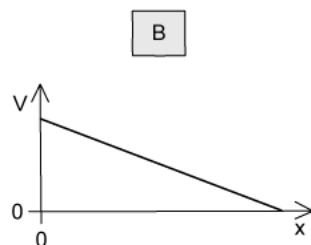
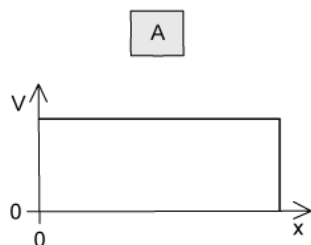
[1 mark]

Question 5

The circular cross-sectional area of a uniform metal wire varies along its length. There is a current in the wire. The narrow end of the wire is at a reference potential of zero.



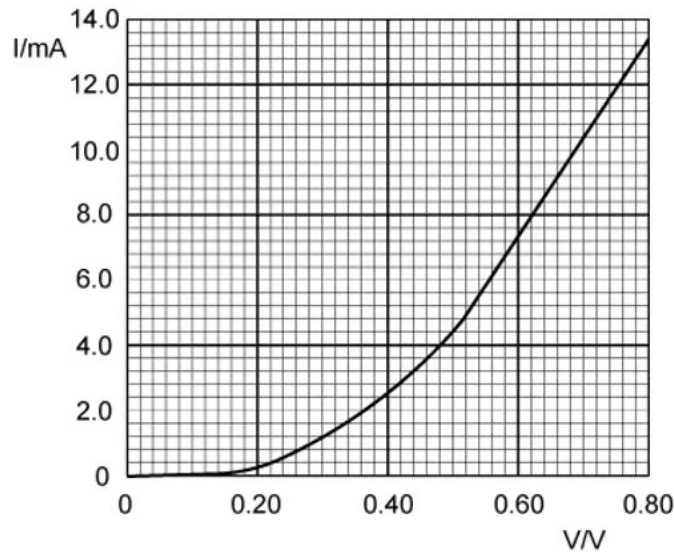
Which graph best represents the variation with distance x along the wire of the potential difference V relative to the reference zero?



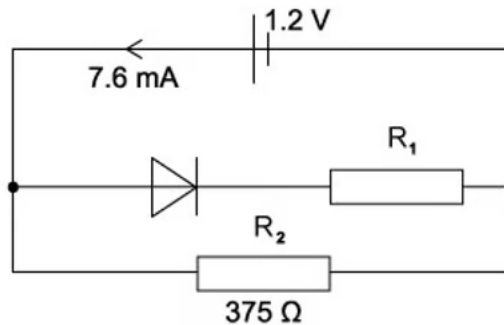
[1 mark]

Question 6

The graph shows the I – V characteristics of a semiconductor diode



A cell of e. m. f. 1.2 V and negligible internal resistance is connected in series to a semiconductor diode and a resistor R_1 , as shown below



A resistor R_2 of resistance $375\ \Omega$ is connected across the cell. The diode has the characteristic shown in the graph. The current supplied by the cell is 7.6 mA.

What is the resistance of R_1 ?

- A. $82\ \Omega$
- B. $114\ \Omega$
- C. $160\ \Omega$
- D. $260\ \Omega$

[1 mark]

Question 7

The graph shows how the current I varies with potential difference V for an electrical component.

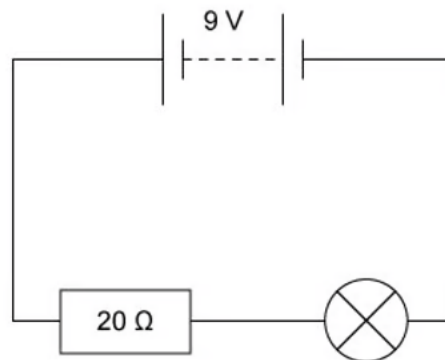
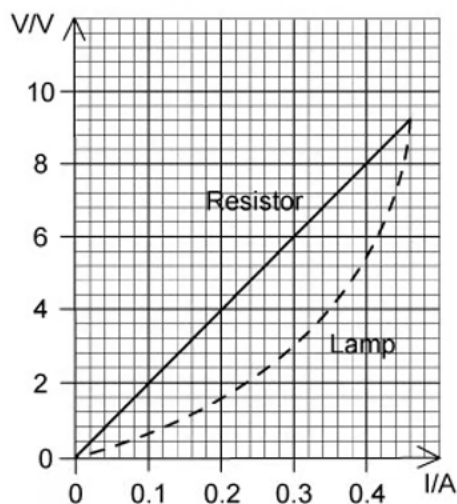
Which row of the table gives the resistance of the component at V_2 and describes how the resistance changes from V_1 to V_2 ?

	Resistance at V_2	Change in resistance from V_1 to V_2
A	$\frac{V_2 - V_1}{I_2 - I_1}$	increases
B	$\frac{V_2 - V_1}{I_2 - I_1}$	decreases
C	$\frac{V_2}{I_2}$	increases
D	$\frac{V_2}{I_2}$	decreases

[1 mark]

Question 8

The graph shows the relationship between potential difference V and current I for a fixed $20\ \Omega$ resistor and a filament lamp.



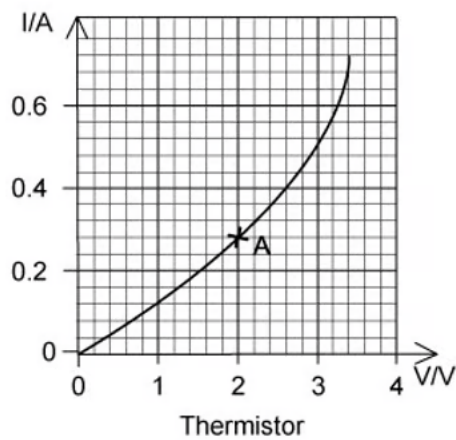
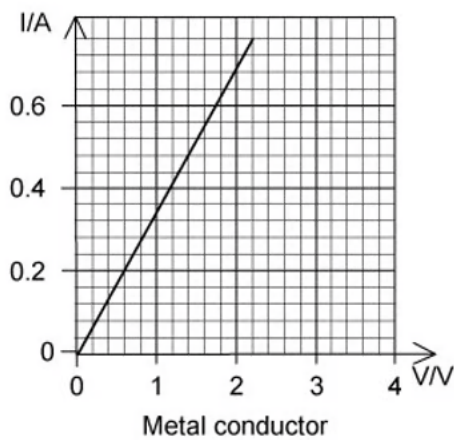
The resistor and lamp are placed in series with a 9 V battery of negligible internal resistance. What is the current in the circuit?

- A. 0.1 A
- B. 0.2 A
- C. 0.3 A
- D. 0.4 A

[1 mark]

Question 9

The graphs show the I - V characteristics for a metal conductor and a thermistor



What is the ratio $\frac{\text{resistance of thermistor at point A}}{\text{resistance of metal conductor}}$?

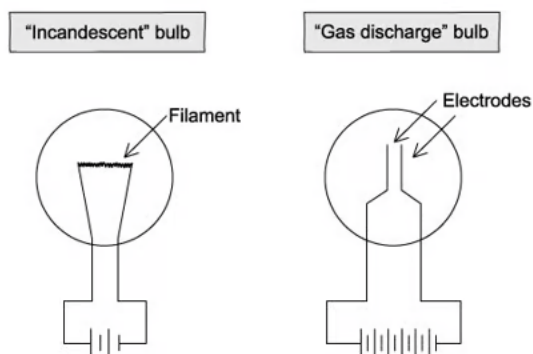
- A. $\frac{2}{5}$
- B. $\frac{1}{2}$
- C. $\frac{2}{1}$
- D. $\frac{5}{2}$

[1 mark]

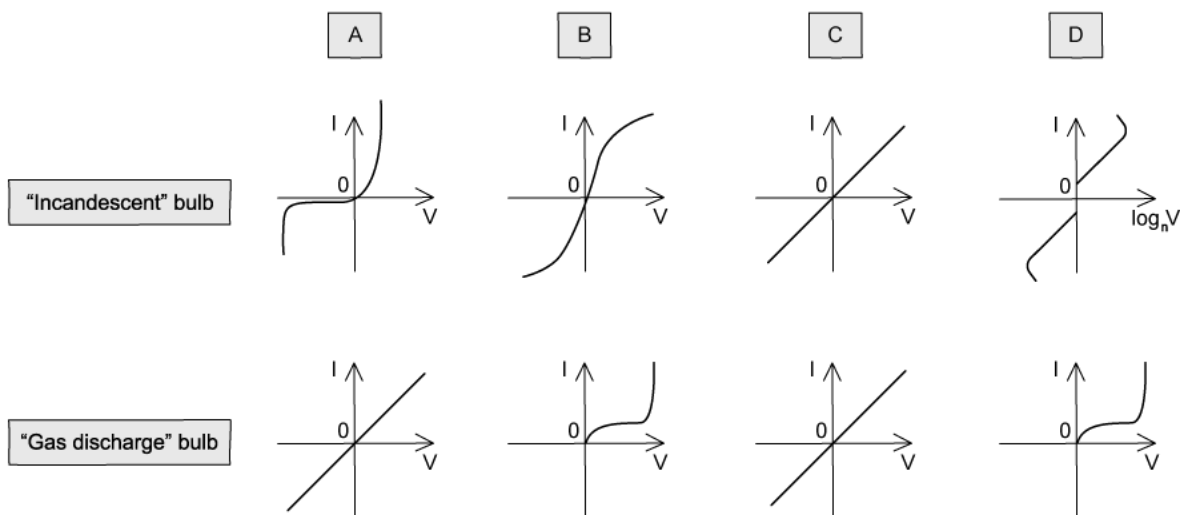
Question 10

One style of light bulb, very different from the "incandescent" design which works on the principle of a super-heated wire filament emitting light, is called a gas discharge tube.

In this design of light bulb, light is produced by the direct excitation of gas molecules as electric current passes between two electrodes.



What will the $I - V$ graphs of these bulbs look like?



[1 mark]

Model Answers: Hard

1

The correct answer is **A** because:

- Ohm's Law states $V = IR$
- For the pump to perform acceptably, we are told the potential difference across it should be at least 218 V
 - So, the potential difference across the cables is $230 - 218 = 12 \text{ V}$
- The current through the electric pump is 0.83 A
- The same current flows through the cable since it is in series with the pump. So, the cable has a total resistance of $R = \frac{230 - 218}{0.83} = 14.5 \Omega$
- From the diagram, it can be seen that the cable is made up of two wires
 - So, the total length of wire in the cable is $2 \times 18 = 36 \text{ m}$
- The wires are not connected to each other - so, they are not in parallel. Instead, one wire connects one terminal of the mains to one terminal of the pump, and the other wire connected the other terminal of the mains to the other terminal of the pump
- This total length of wire corresponds to a total resistance of 14.5Ω
 - Therefore, the resistance per meter is equal to $= \frac{14.5}{36} = 0.4 \Omega \text{ m}^{-1}$

2

The correct answer is **B** because:

- Ohm's Law states that $V = IR$, or $R = \frac{V}{I}$
- This means the slope of the $I - V$ graph is equal to $\frac{1}{R}$
- Therefore, as current I increases, R increases gradually as the slope decreases
- Electrical power is equal to $P = VI = (IR)I = I^2R$, or $R = \frac{P}{I^2}$
- The slope of a $P - I^2$ graph is equal to R
- Therefore, from the graph of $P - I^2$, as R increases the slope of $P - I^2$ increases as I increases

- Hence, the graph of $P - I^2$ will be the inverse of the $I - V$ graph

3

The correct answer is **D** because:

- In the first test, when all the switches are open, the meter reads infinity
 - This is obvious since the circuit is incomplete and no current flows
- In the second test, where only S_1 is closed, we can see that lamp A is not faulty
 - This is because the ohm-meter reads 15Ω , proving that a current is passing through component A
- In the third test where S_1 is open, S_2 is closed and S_3 is open, the meter reads a resistance of 30Ω , so, this test proves that lamp B is also not faulty
 - This is because components A and B are now connected in series, giving a total resistance of $15 + 15 = 30\Omega$
- In the last test, where only S_1 is open, the resistance reading is 15Ω
- This proves that lamp C is shorted, meaning the current passes through as if there was no lamp
 - In this case, it's as if a wire of negligible resistance is connected in parallel with lamp B, and since Ohm's Law, $I \propto \frac{1}{R}$ the current is larger when the resistance is smaller
 - So, all current would flow in that wire instead of through lamp B (since lamp B has some resistance)
 - Now, the only resistance encountered by the current is due to lamp A only, so, that's why the reading on the ohm-meter is 15Ω

Exam Tip: Remember current flow in a circuit will always take the route with the least resistance. If S_1 and S_2 were closed, and S_3 was opened, the ohm-meter would again read 15Ω since all the current would pass through S_1 (which has no resistance) instead of passing through lamp B (which has some resistance) and therefore, the current would then flow through lamp A (which is the only lamp that causes a resistance in this case)

4

The correct answer is **C** because:

- Ohm's Law states that $V = IR$, or $R = \frac{V}{I}$
- This means the slope of the $I - V$ graph is equal to $\frac{1}{R}$
- The $I - V$ graph consists of two sections
 - The first section at lower voltages has constant resistance because it is a linear $I - V$ relationship
 - The second half at higher voltages has increasing resistance because the gradient is decreasing
- Therefore, in an $R - V$ graph, initially the line would be horizontal as $\frac{V}{I}$ is constant, then, the line will begin to curve upwards as R increases
 - Option C corresponds best to this description

A is incorrect because the resistance does not stay constant, or the $I - V$ graph would be completely straight

B is incorrect because even though the first section of this graph is correct, the second half shows the resistance is decreasing in the second half, which is incorrect

D is incorrect because the first half of the $I - V$ graph shows the component has an initial resistance which is constant, it does not begin at zero and it does not increase linearly

5

The correct answer is **C** because:

- Ohm's Law states $V = IR$
 - In this case, the current is constant along the wire, only the resistance changes and this causes a proportional change in the potential difference
- Resistance of wire is equal to $R = \frac{\rho L}{A}$
- When the length of wire L is at a maximum distance from the zero potential reference, resistance R will also be at a maximum and hence, potential difference V too (since $R \propto L \propto V$)
 - So, as distance x increases, the length L decreases, and so, R and V also decrease
- Now, the diameter of the wire is falling linearly as x increases, but the area of cross-section A is not falling linearly as $A = \pi \left(\frac{d}{2}\right)^2$ where d is the diameter

- Since resistance R is inversely proportional to cross-sectional area A ($R \propto \frac{1}{A}$) it is also inversely proportional to d^2 (when d is small, R is large)
- Therefore, there is a greater percentage fall (in resistance, and hence, potential difference) per unit length at the narrow end than at the wide end, meaning the decrease is not linear, but depends on d^2
- $\frac{V}{x}$ This means that potential difference per unit length (this is represented by the gradient of the graphs shown which equal to) is less at the wide end than at the narrow end (the gradient at the highest value of x should be greater than the gradient at $x = 0$ and the gradient keeps on increasing as x increases)

A is incorrect because V decreases with distance, whereas this graph shows it stays constant

B is incorrect because V depends on d^2 and therefore does not decrease linearly

D is incorrect because the gradient is highest at $x = 0$ and gradually increases

6

The correct answer is **C** because:

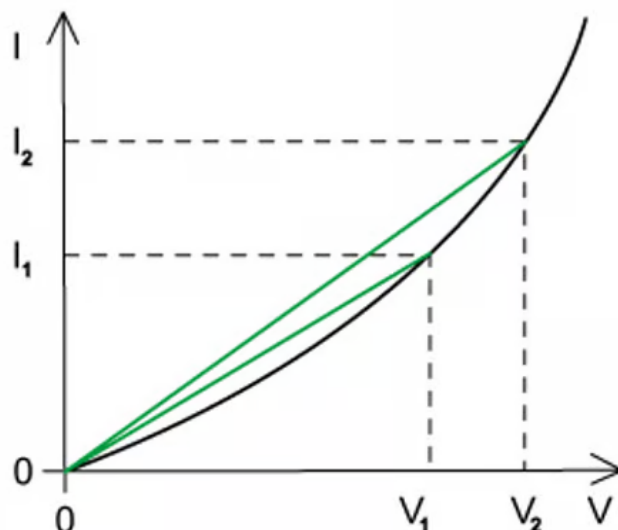
- The p.d. across each loop is equal to the e. m. f. in the circuit
 - Ohm's Law is $I = \frac{V}{R}$
- So, the current across the 375Ω resistor is $\frac{1.2}{375} = 3.2 \times 10^{-3} \text{ A} = 3.2 \text{ mA}$
 - This means the current through the semiconductor diode and R_1 is $7.6 - 3.2 = 4.4 \text{ mA}$
- To find the p. d. across the semiconductor diode we need to use the graph
 - For a current of 4.4 mA , the p. d. from graph is 0.5 V
- So, p. d. across $R_1 = 1.2 - 0.5 = 0.7 \text{ V}$
 - Resistance is equal to $R = \frac{V}{I} = \frac{0.7}{4.4 \times 10^{-3}} = 159 \Omega$
- Therefore, the resistance across R_1 is 160Ω

7

The correct answer is **D** because:

- Ohm's Law states that $V = IR$, or $R = \frac{V}{I}$
- The resistance at V_2 is simply found by reading the values of current and p. d. from the axes, **not** from taking the tangent between the two points
- This is because resistance is defined by $\frac{V}{I}$ **not** $\frac{\Delta V}{\Delta I}$
 - So, the resistance at V_2 is equal to $\frac{V_2}{I_2}$
- To see how the resistance changes from V_1 to V_2 we have to consider the tangents from the **origin** to the two points (not the tangents on the curve)

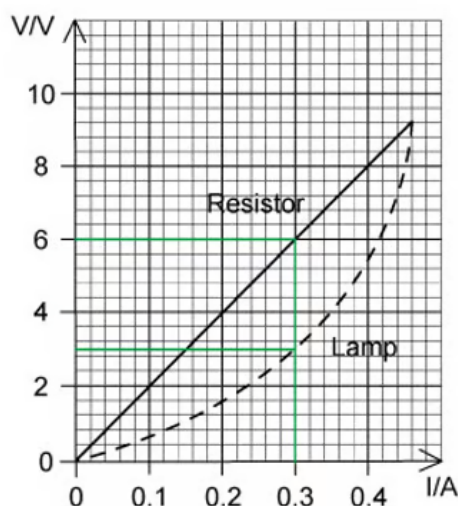
- To see how the resistance changes from V_1 to V_2 we have to consider the tangents from the **origin** to the two points (not the tangents on the curve)
- From Ohm's Law, we know the slope of the $I - V$ graph is equal to $\frac{1}{R}$
 - So, the steeper the gradient, the smaller the value of resistance
- The gradient gets steeper from V_1 to V_2 , meaning the resistance is decreasing



8

The correct answer is **C** because:

- In this question, we must look at the values of potential difference (p.d.) of each component at each current to determine which two values of p. d. add up to make 9 V
- Consider option C, at 0.3 A:
 - The p. d. of the resistor is 6 V
 - The p. d. of the lamp is 3 V
- In a series circuit, the e. m. f from the battery is split between the components, so we just add them together
 - So, the total p. d. of the components is $6 + 3 = 9$ V
- This is the e. m. f of the battery, so the current is 0.3 A



A is incorrect because p.d. of the resistor = 2 V, p.d. of the lamp = 0.6 V

Total p.d. = 2.6 V

B is incorrect because p.d. of the resistor = 4 V, p.d. of the lamp = 1.6 V

Total p.d. = 5.6 V

D is incorrect because p.d. of the resistor = 8 V, p.d. of the lamp = 5.4 V

Total p.d. = 13.4 V

9

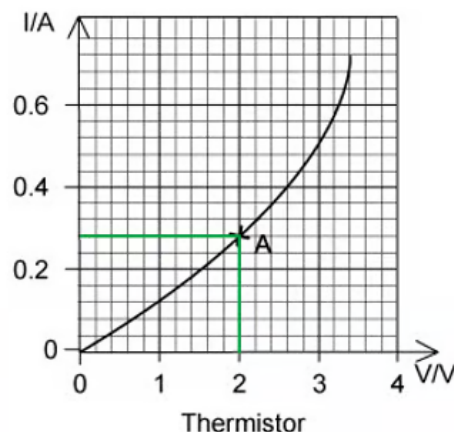
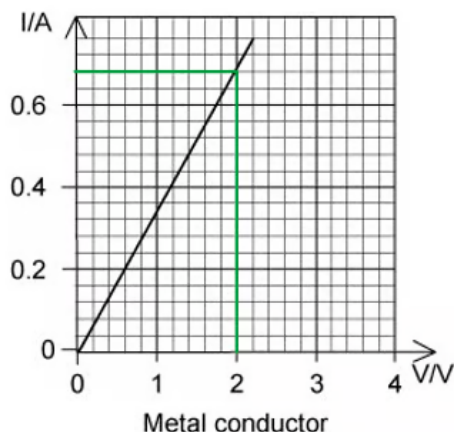
The correct answer is **D** because:

- Ohm's Law states that $V = IR$, or $R = \frac{V}{I}$
- For the metal conductor:
 - At $V = 2$ V, the current is $I = 0.68$ A
- So, the resistance of the metal conductor is:
 - $R_M = \frac{V}{I} = \frac{2}{0.68} = \frac{50}{17} = 3\Omega$
- For the thermistor, at point A:
 - At $V = 2$ V, the current is $I = 0.27$ A
- The resistance at point A is simply found by reading the values of current and p.d. from the axes, **not** from taking the tangent of this point on the curve, so the resistance of the thermistor is:

$$\circ R_T = \frac{V}{I} = \frac{2}{0.27} = \frac{200}{27} = 7.4 \, \Omega$$

- So, the ratio is given by :

$$\circ \frac{R_T}{R_M} = \left(\frac{200}{27} \right) \div \left(\frac{50}{17} \right) = \left(\frac{5}{2} \right)$$

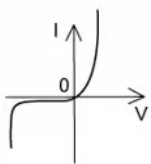


10

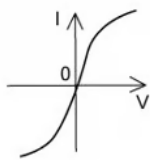
The correct answer is **B** because:

- An incandescent bulb is equivalent to a filament lamp, in which the resistance increases as V increases
- You are not expected to know what the graph of a gas discharge bulb looks like, but just to realize it is non-Ohmic
- A device is said to be non-Ohmic if the resistance across it is not constant, but changes in a way that depends on the voltage across it
- In this case, the I - V graph is not a straight line but has some curvy shape. Devices such as this do not have a constant value of resistance; the resistance is called dynamic resistance because it is constantly changing as V and I change

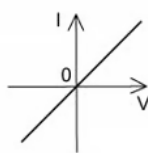
The I - V graphs are for the following components:



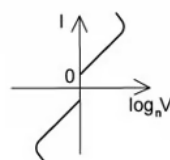
Semiconductor diode



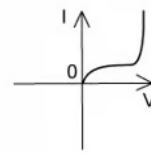
Filament lamp



Ohmic conductor



Thermistor



Gas discharge tube

Exam Tip: Note the graph for the thermistor is plotted as I against $\log_e V$, making it